NAREN KARTHIK

Electronics & Communication Engineering Student
IoT, Networking & Systems

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in IoT systems, Linux-based servers, networking fundamentals, and security-focused projects using ESP32 and Python. Seeking internship or entry-level opportunities to apply practical skills and gain industry exposure.

EDUCATION

ANNA UNIVERSITY

Bachelor of Engineering – Electronics & Communication Engineering
CGPA: 7.65 | 2023 – 2027

TECHNICAL SKILLS

Core Skills
- Computer Networks (TCP/IP, OSI, Routing Basics)
- IoT System Design & Sensor Integration
- Networking & Security Fundamentals
- Embedded Systems

Programming & Tools
- Python (Basics, NLP Projects)
- C / Embedded C (ESP32, Arduino)
- Git & GitHub (Basics)
- HTML & CSS (Basics)

Hardware & Platforms
- ESP32, Arduino
- Sensors: DHT11, Ultrasonic, Soil Moisture, Rain
- LoRa (SX1278), Relay Modules
- Linux (CLI)

PROJECTS & PRACTICAL EXPERIENCE

Self-Hosted Home Cloud & Web Server (antiX Linux) **2025-DEC**
Built a self-hosted cloud and web server on an old laptop using Linux.
Configured persistent services, firewall rules, and secure remote access via Tailscale.
Debugged service failures, reboot persistence, and network reachability issues.
Tech: Linux (antiX), Networking, SSH, UFW, Tailscale

AI-Powered Network Intrusion Detection System (NIDS) **2025-AUG**
Built a real-time network intrusion detection system using Python and Scapy.
Trained ML models on NSL-KDD and CICIDS2017 datasets for attack detection.
Visualized traffic and alerts through an interactive dashboard.
Tech: Python, Scapy, Machine Learning, Network Security

Environmental Stress Index (ESI) – ESP32 Based IoT System **2026-JAN**
Built an ESP32-based environmental monitoring system with a live web dashboard.
Combined multiple sensors into a single Environmental Stress Index.
Performed sensor calibration and ADC tuning for reliable readings.
Tech: ESP32, Embedded C, IoT, Sensors

E-commerce Customer Support Chatbot **2025-MAY**
Built an intent-based customer support chatbot using NLP and machine learning.
Implemented JSON-based intent classification for automated responses.
Tech: Python, TensorFlow, NLTK

RIP-Based Multi-Router Network Simulation **2025-APR**
Simulated a 6-router RIP v2 topology with DNS and HTTP services.
Tech: Cisco Packet Tracer, RIP, Networking

CERTIFICATIONS & TRAININGS

Oracle Certified Foundations Associate - Oracle University
Cybersecurity Analyst Job Simulation - Tata Consultancy Services (Forage)
Industrial IoT & Industry 4.0 - Tamil Nadu Skill Development Corporation
Data Analytics & Visualization Job Simulation - Accenture (Forage)

GitHub: github.com/Narenkarthik-07 **| Linkedin:** linkedin.com/in/narenkarthik17